

PRODUCT INFORMATION

Flow Master

Accurate, Stable, Handy Flow Generator

Customized to the specified pressure and flow.
Use as a standard for testing and comparing flowmeters.

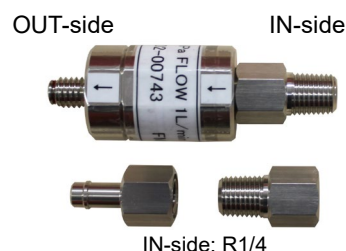
- Check the sensitivity of flow testers easily and reliably.
- Maintains the same flow for a long period of time.
- Use as a flow master for measuring deviation flow.
(Test for ventilation and bore diameter of holes)
- Can be connected to the CAL port of Cosmo's flow tester.
- Can be used in Flow Check (C-CHK), (F-CHK) of AF series.
- Joints for connecting to the piping available.
Easy connection and disconnection with a plug.
- Traceability certificate available upon request.
- ISO/IEC17025 calibration available.

Customized to the desired flow. Use it as a reference instrument.

FM-2A

Flow Master

Accuracy (Difference from the required flow) : Within $\pm 10\%$



FM-2A

Model: FM-2A (A)(B,C)

- A** Adapter diameter: IN-side: R1/4, R1/2
OUT-side: R1/4, $\phi 8$ tube joint
- B** Required flow: mL/min L/min
Up to 999 mL/min (in units of 0.1 mL/min)
Up to 20 L/min (in units of 0.1 L/min)
Beyond that, specify in units of up to 1 L/min.
- C** Pressure: FM-2A kPa
FM-2A: For less than 200 kPa: In units of 0.1 kPa.
For others: In units of 1 kPa.

- **Main Body & Accessories** FM-2A
Accessories: IN-side and OUT-side joints (1 set),
O-ring (3 pcs.)

Specifications

[R2]: (IN-side: R1/4, OUT-side: R1/4)
[R21]: (IN-side: R1/4, OUT-side: $\phi 8$ tube joint)
[R3]: (IN-side: R1/2, OUT-side: R1/4)
[R31]: (IN-side: R1/2, OUT-side: $\phi 8$ tube joint)

Example:

Adapter diameter: IN-side: R1/2
OUT-side: $\phi 8$ tube joint
Flow 4 L/min, Pressure 150 kPa
Model: FM-2A(R31)(4L/min.150kPa)

Flow	Pressure	
50 mL/min to 10 L/min	1 to 9.9 kPa	Pressure
301 mL/min to 20 L/min	10 to 20.9 kPa	
301 mL/min to 30 L/min	21 to 50.9 kPa	
301 mL/min to 40 L/min	51 to 99.9 kPa	
501 mL/min to 55 L/min	100 to 150.9 kPa	
501 mL/min to 75 L/min	151 to 200 kPa	
501 mL/min to 80 L/min	201 to 250 kPa	
501 mL/min to 90 L/min	251 to 300 kPa	
501 mL/min to 95 L/min	301 to 350 kPa	
501 mL/min to 100 L/min	351 to 400 kPa	
50 mL/min to 10 L/min	-1 to -9.9 kPa	Vacuum
101 mL/min to 10 L/min	-10 to -20.9 kPa	
101 mL/min to 20 L/min	-21 to -49.9 kPa	
201 mL/min to 20 L/min	-50 to -89.9 kPa	

About Flow

For FM-2A, the flow is indicated by the "Conversion Flow".

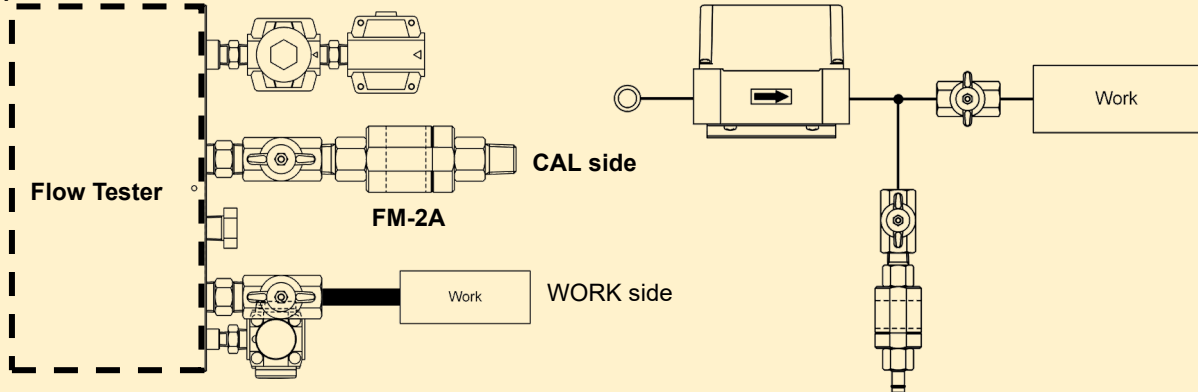
- Conversion Flow: Flow of **Flow Master** measured and converted at calibration environment. (The flow varies depending on the condition of the calibration environment.)

How to Use

FM-2A is a standard flowmeter with a fixed pressure and flow. Unlike float-type flowmeters, there is no need to determine the flow using a regulator valve, and no trouble such as damage to the float due to inrush pressure.

Since standard flow rate can be set, it is ideal for accuracy check.

• Example of Flow Master Installation



(1) When connecting the Flow Master to Flow Tester

Connect the Flow Master to the **CAL** port.

(*1, *2)

*1 For mass flow sensor: R1/2(R3,R31)

*2 For laminar flow sensor:

10 to 200 mL/min: R1/4(R2,R21)

500 mL/min or more: R1/2(R3,R31)

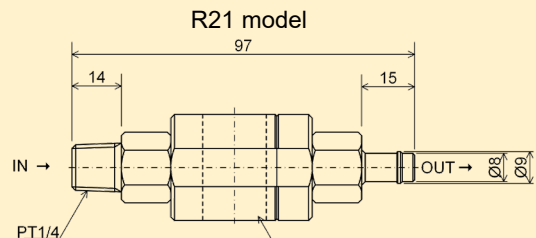
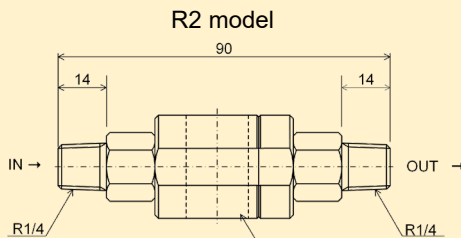
(2) When connecting the Flow Master to the piping

Connect as shown above.

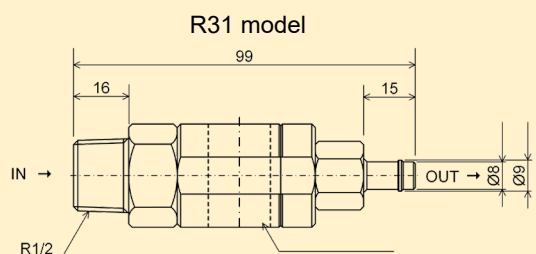
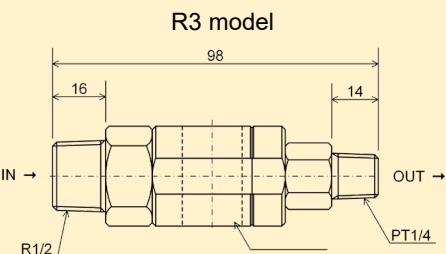
Attach a shutoff valve in front of the Flow Master. It will always be possible to test with the Flow Master kept connected. (Open when in use, Close when not in use)

External Appearance

R2 Series Male side: R1/4



R3 Series Male side: R1/2



* The contents of this Product Information are as of May 2022. The specifications are subject to change without prior notice.

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